

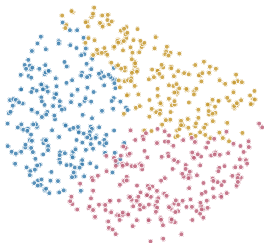
Figure 1 | Threadfin connects receptor identity to B-cell state relationships

Integrating paired scRNA-seq and scBCR-seq to resolve clone-state organisation.

Paired single cells

scRNA-seq

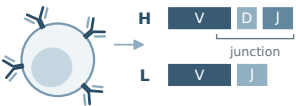
Expression state



One dot = one cell

scBCR-seq

Heavy + light-chain sequences



B cell Paired by cell barcode



Sequence families within donors

Identify GC selection-associated nodes

One point = one receptor family

Return to GC cycling?

Candidate direction

Memory-like

Selection-associated GC node

GC cycling

Output-like states

Resolve cycling-to-output clone gradients

Captured cells

• 5 • 20 • 80

Cycling

Output-like

